

Computer Architecture- Quiz #5-C

Name: Aabia Ali

Roll#: 23i-0704

Section: D

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 15

Propagation delay of a TTL based Full adder is approximately 10 ns. Calculate delay for a 16-bit ripple carry adder based on TTL based full adders? (2)

$$= 16 \times t_{FA}$$

$$= 16 \times 10$$

$$= 160 \text{ ns}$$

Consider Carry-Lookahead Adder and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals. (3)

$$G_i = A_i \cdot B_i$$

$$, P_i = A_i + B_i$$

$$C_i = G_i + P_i \cdot C_{i-1}$$

Calculate Column propagate P_i and generate signals G_i for each column if $A=11001$, $B=10111$

$\begin{array}{r} 11001 \\ 10111 \\ \hline \end{array}$	$\begin{array}{r} 11001 \\ 10111 \\ \hline \end{array}$
$\begin{array}{r} G_i \quad 10001 \\ \hline \end{array}$	$\begin{array}{r} P_i \quad 11111 \\ \hline \end{array}$
$\begin{array}{cccccc} G_4 & G_3 & G_2 & G_1 & G_0 & \\ \hline \end{array}$	$\begin{array}{cccccc} P_4 & P_3 & P_2 & P_1 & P_0 & \\ \hline \end{array}$

For a 4-bit carry look-ahead adder, derive the expressions for the block propagate $P_{3:0}$ and block generate $G_{3:0}$ signals. Also, express the carry-out C_3 in terms of these block propagate and generate signals. (5)

$$P_{3:0} = P_3 \cdot P_2 \cdot P_1 \cdot P_0$$

$$G_{3:0} = G_3 + G_2 \cdot P_3 + G_1 \cdot P_2 \cdot P_3 + G_0 \cdot P_1 \cdot P_2 \cdot P_3$$

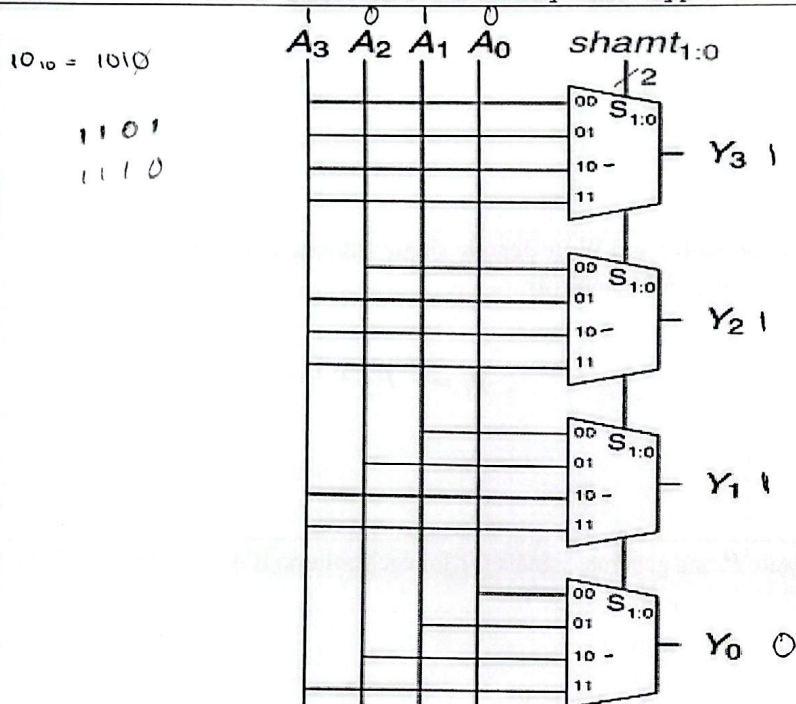
$$C_3 = G_{3:0} + P_{3:0} C_{-1}$$

CLO 2:

Question 2

Marks 5

Consider the given circuit. Evaluate its operation for input $A = 10$ and shift amount $shamt = 2$. Determine the final output $Y_{0:3}$ value in decimal form and identify the type of the circuit.

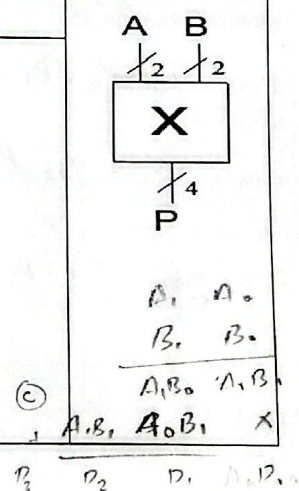
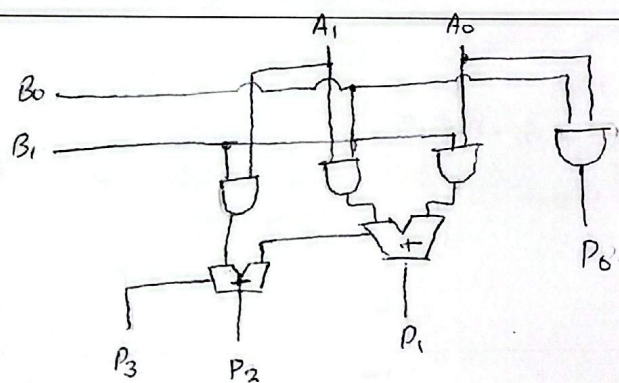


CLO 2:

Question 3

Marks 5

Design a 2 x 2 multiplier?



Computer Architecture- Quiz #5-C

Name: Fatima Sohail

Roll#: 1230633

Section: D

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 15

Propagation delay of a TTL based Full adder is approximately 10 ns. Calculate delay for a 16-bit ripple carry adder based on TTL based full adders? (2)

$$\begin{aligned} d_{\text{Tr}} &= N \times T_{\text{FA}} \\ &= 16 \times 10 \\ &= 160 \text{ ns} \end{aligned}$$

Consider Carry-Lookahead Adder and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals. (3)

$$G_i = A_i \cdot B_i$$

$$P_i = A_i \oplus B_i$$

$$C_i = G_i + P_i \cdot C_{i-1}$$

Calculate Column propagate P_i and generate signals G_i for each column if $A = 11001$, $B = 10111$

$$\begin{array}{r} 11001 \\ 10111 \\ \hline P = 11111 \\ G = 10001 \end{array}$$

$$\begin{array}{ll} P_0 = 1 & G_0 = 1 \\ P_1 = 1 & G_1 = 0 \\ P_2 = 1 & G_2 = 0 \\ P_3 = 1 & G_3 = 1 \end{array}$$

For a 4-bit carry look-ahead adder, derive the expressions for the block propagate $P_{3:0}$ and block generate $G_{3:0}$ signals. Also, express the carry-out C_3 in terms of these block propagate and generate signals. (5)

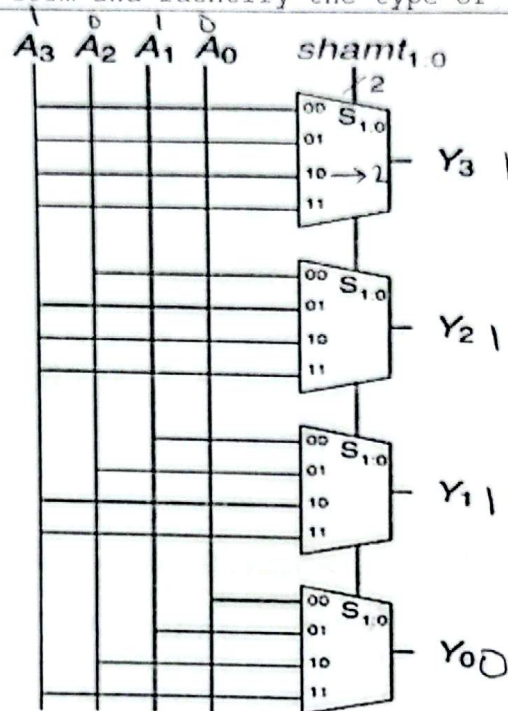
$$P_{3:0} = P_3 \cdot P_2 \cdot P_1 \cdot P_0$$

$$G_{3:0} = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0 + P_3 P_2 P_1 P_0$$

$$C_3 = P(G_{3:0}) + P_{3:0} C_2$$

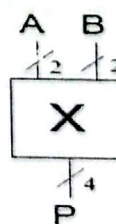
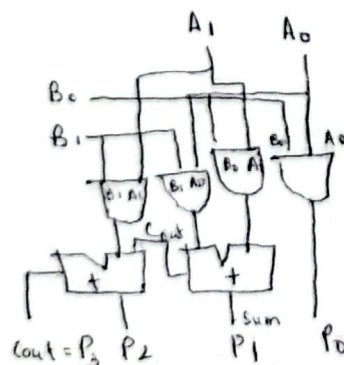
$$\begin{aligned} & \xrightarrow{A_3+B_3} (A_3+B_3)(A_2+B_2) \cdot (A_1+B_1) \cdot (A_0+B_0) \\ & \xrightarrow{A_3B_3 + (A_3+B_3)(A_2B_2) + (A_3+B_3)(A_2+B_2)(A_1B_1) + (A_3+B_3)(A_2+B_2)(A_1+B_1)(A_0B_0) + (A_3+B_3)(A_2+B_2)(A_1+B_1)(A_0+B_0)} \\ & \xrightarrow{A_3B_3 + (A_3+B_3)(A_2B_2) + (A_3+B_3)(A_2+B_2)(A_1B_1) + (A_3+B_3)(A_2+B_2)(A_1+B_1)(A_0B_0) + (A_3+B_3)(A_2+B_2)(A_1+B_1)(A_0+B_0)} \end{aligned}$$

Marks 5



Name: shift right arithmetic

Marks 5



Computer Architecture- Quiz #5-C

Name: Menawil

Roll#: 231-0546

Section: D

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 15

Propagation delay of a TTL based Full adder is approximately 10 ns. Calculate delay for a 16-bit ripple carry adder based on TTL based full adders? (2)

$$10 \times 16 = 160 \text{ ns}$$

Consider Carry-Lookahead Adder and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals. (3)

$$G_i = A_i \cdot B_i$$

$$P_i = A_i + B_i$$

$$C_i = G_i + P_i C_{in}$$

Calculate Column propagate P_i and generate signals G_i for each column if $A=11001$, $B=10111$

$$\begin{array}{r} 11001 \\ 10111 \\ \hline L \quad 10001 \\ P \quad 11111 \end{array}$$

For a 4-bit carry look-ahead adder, derive the expressions for the block propagate $P_{3:0}$ and block generate $G_{3:0}$ signals. Also, express the carry-out C_3 in terms of these block propagate and generate signals. (5)

$$P_{3:0} = \cancel{A_3} \cdot P_3 \cdot P_2 \cdot P_1 \cdot P_0$$

$$G_{3:0} = G_3 + G_2 P_3 + G_1 P_2 P_3 + G_0 P_1 P_2 P_3$$

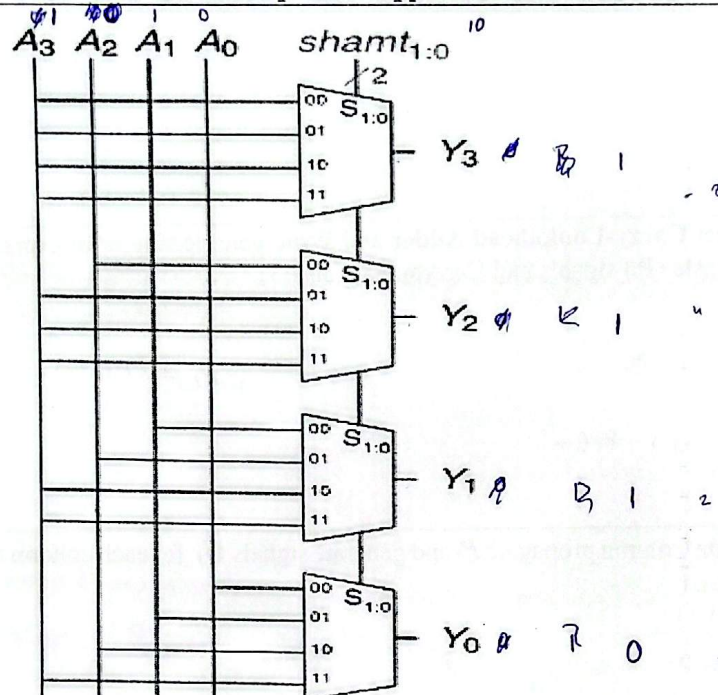
$$C_3 = G_{3:0} + P_{3:0} C_{in}$$

CLO 2:

Question 2

Marks 5

Consider the given circuit. Evaluate its operation for input $A = 10$ and shift amount $shamt = 2$. Determine the final output $Y_{0:3}$ value in decimal form and identify the type of the circuit.



$Y_{0:3} = 0101$

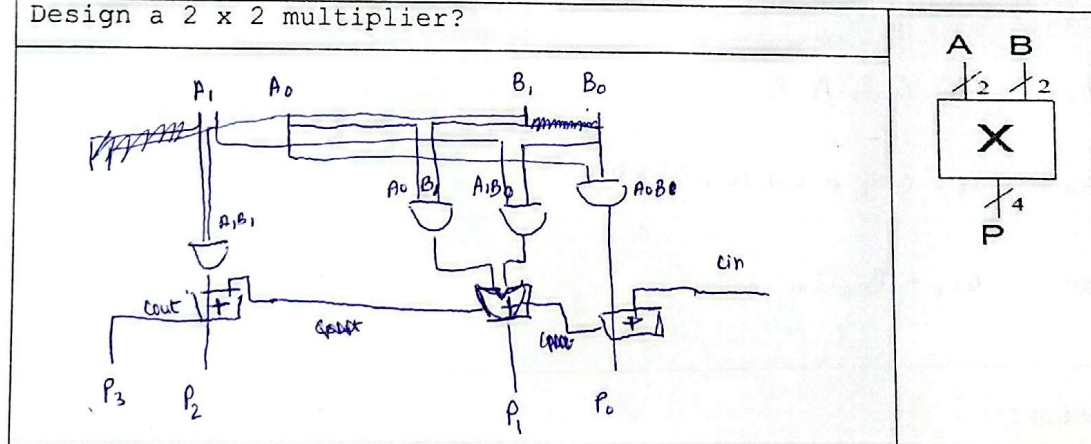
Name: Arithmetic Right Shifter

CLO 2:

Question 3

Marks 5

Design a 2 x 2 multiplier?



$$\begin{array}{r}
 \begin{array}{r}
 0110 \\
 0101 \\
 1111 \\
 \hline
 10100
 \end{array}
 \times
 \begin{array}{r}
 A_1 A_0 \\
 B_1 B_0 \\
 \hline
 A_1 B_0 A_0 B_0 \\
 A_1 B_1 A_0 B_1 \\
 \hline
 P_3 P_2 P_1 P_0
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 10 \\
 \times 10 \\
 \hline
 00 \\
 10 \\
 \hline
 100
 \end{array}$$

Computer Architecture- Quiz #5-C

Name: ABDUL GHUFRAN Roll#: 23i-0576

Section: C

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 15

Propagation delay of a TTL based Full adder is approximately 10 ns. Calculate delay for a 16-bit ripple carry adder based on TTL based full adders? (2)

$$= 10 \text{ ns} \times 16$$

$$= 160 \text{ ns}$$

Consider Carry-Lookahead Adder and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals. (3)

$$G_i = A_i B_i$$

$$P_i = A_i + B_i$$

$$C_i = G_i + P_i \cdot C_{i-1}$$

Calculate Column propagate P_i and generate signals G_i for each column if $A = 11001$, $B = 10111$

$$P_0 = 1, P_1 = 1, P_2 = 1, P_3 = 1, P_4 =$$

$$G_0 = 1, G_1 = 0, G_2 = 0, G_3 = 0, G_4 = 1$$

11001	11001
10111	10111
10001	11111
AND	OR

For a 4-bit carry look-ahead adder, derive the expressions for the block propagate $P_{3:0}$ and block generate $G_{3:0}$ signals. Also, express the carry-out C_3 in terms of these block propagate and generate signals. (5)

$$P_{3:0} = A_3 + B_3, A_2 + B_2, A_1 + B_1, A_0 + B_0$$

$$G_{3:0} = A_3 B_3, A_2 B_2, A_1 B_1, A_0 B_0$$

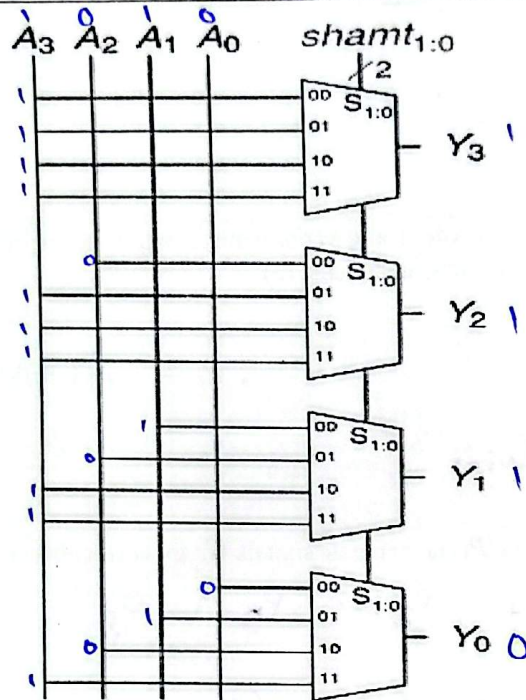
$$C_3 = G_3 + G_2 \cdot P_3 + G_1 \cdot P_2 \cdot P_3 + G_0 \cdot P_1 \cdot P_2 \cdot P_3$$

CLO 2:

Question 2

Marks 5

Consider the given circuit. Evaluate its operation for input $A = 10$ and shift amount $shamt = 2$. Determine the final output $Y_{0:3}$ value in decimal form and identify the type of the circuit.



$Y_{0:3} = 1110$

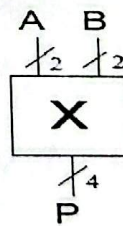
Name: Shift Right Arithmetic

CLO 2:

Question 3

Marks 5

Design a 2×2 multiplier?



Computer Architecture- Quiz #5-C

Name: Abdul Rauf

Roll#: 237-0591

Section: C

Total Time: 20 minutes

Total Marks: 20

7th April, 2026

CLO 2:

Question 1

Marks 15

Propagation delay of a TTL based Full adder is approximately 10 ns. Calculate delay for a 16-bit ripple carry adder based on TTL based full adders? (2)

$$\begin{aligned} \text{Delay} &= N \times t_{FA} \\ &= 16 \times 10 \times 10^{-9} \\ &= 1.6 \times 10^{-7} \text{ s.} \end{aligned}$$

Consider Carry-Lookahead Adder and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals. (3)

Let $i = 3$

$$G_i = G_i + G_{i-1}P_i + P_iP_{i-1}G_{i-2} \dots, P_i = P_{i-1} \cdot P_{i-2} \dots P_0.$$

$$C_i = G_i + G_{i-1}P_{i-1} \dots = P_{i-1} \dots P_0.$$

$$C_i = G_{i:0} + P_{i:0} \cdot C_{n-1}$$

Calculate Column propagate P_i and generate signals G_i for each column if $A = 11001$, $B = 10111$

$$\begin{aligned} A &= 11001 \\ B &= 10111 \\ P_i &= A_i \oplus B_i \\ G_i &= A_i B_i \end{aligned}$$

$$\begin{aligned} A &= 11001 \\ B &= 10111 \\ P_i &= 11111 \\ G_i &= 10001 \end{aligned}$$

For a 4-bit carry look-ahead adder, derive the expressions for the block propagate $P_{3:0}$ and block generate $G_{3:0}$ signals. Also, express the carry-out C_3 in terms of these block propagate and generate signals. (5)

$$P_{3:0} = P_3 \cdot P_2 \cdot P_1 \cdot P_0.$$

$$G_{3:0} = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0.$$

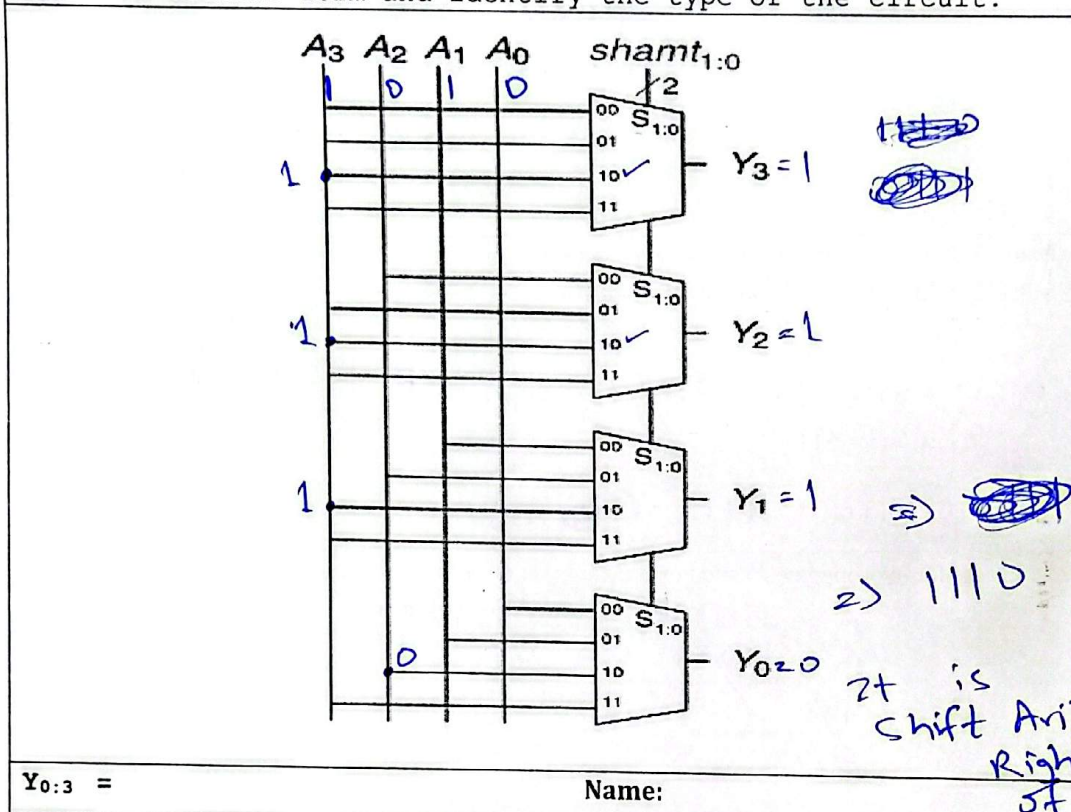
$$C_3 = G_{3:0} + P_{3:0}(C_{-1})$$

CLO 2:

Question 2

Marks 5

Consider the given circuit. Evaluate its operation for input $A = 10$ and shift amount $shamt = 2$. Determine the final output $Y_{0:3}$ value in decimal form and identify the type of the circuit.



CLO 2:

Question 3

Marks 5

Design a 2 x 2 multiplier?

Handwritten multiplication:

```

A1 A0
10
X 01 B0
-----
11
00x
-----
011

```

